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INDUSTRIAL CARBON TRACKING AND EMISSION

IN

SUSTAINABLE ENERGY & SAFTY

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AI/ML for Sustainable Manufacturing

Detecting CO₂ Emissions in Factories

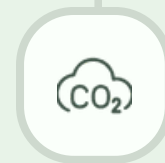
Using existing three-phase energy meters to estimate and optimize CO₂ emissions without installing new sensors.



Concept: Turning Meters into CO₂ Detectors

Existing Infrastructure

Factories use three-phase energy meters (A, B, C) for electricity usage.



Indirect Emissions

CO₂ is produced based on electricity source (coal, gas, renewable).



AI/ML Estimation

Model predicts CO₂ from meter readings (V, A, PF, kWh) and optional production logs.

This approach detects hidden inefficiencies and abnormal emissions in both small and large factories.

Workflow of Carbon Tracking

Three-Phase Energy Meter Readings

This is where raw data originates.

Preprocessing & Feature Extraction

Clean and prepare the raw meter data for the AI model.

AI/ML Model

Use machine learning to **predict CO₂ emissions** from meter readings and identify **unusual energy patterns**

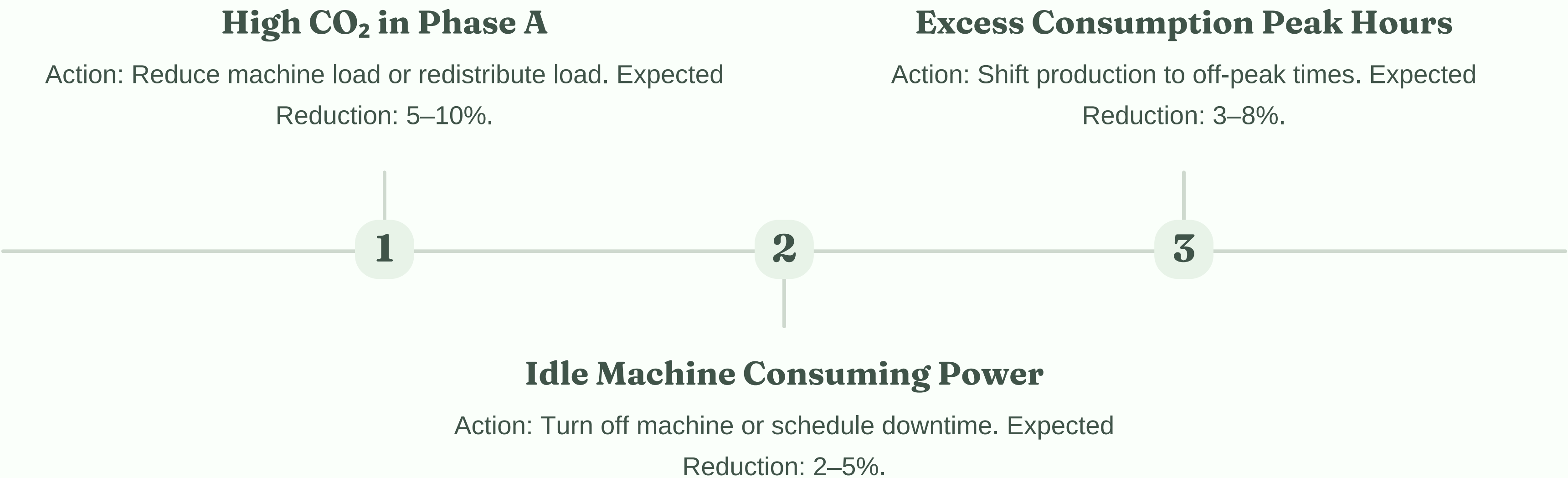
Alerts & Optimization Suggestions

Translate model insights into **actionable feedback** for factory operators.



Actionable Optimization Suggestions

The system generates alerts and suggested actions to reduce emissions and improve efficiency.



📄 Key Advantages: No extra sensors needed, real-time monitoring, detects hidden inefficiencies, and is scalable.

The Dry Ice Carbon Cycle

Dry ice is solid CO₂ used across seven major industries, from food preservation to medical transport.

Food & Beverage



Medical & Pharma



Research



Manufacturing



Logistics



Entertainment



Converting CO₂ to Dry Ice: The Process

This process supports carbon recycling and creates useful products, but faces challenges like high energy use and setup costs.

1. CO₂ Capture

Capture CO₂ from factory exhaust using chemical filters (e.g., amine scrubbing).

2. Purification & Compression

Clean gas to remove impurities, then compress to a liquid state.

3. Cooling & Expansion

Rapid cooling causes part of the liquid CO₂ to freeze into dry ice snow.

4. Shaping

Dry ice snow is pressed into blocks, pellets, or slices for industrial use.



The Challenge: Not Permanent Carbon Storage

Dry ice sublimates back into CO₂, delaying, but not avoiding, atmospheric release. Permanent solutions are required.

Carbon Capture & Storage (CCS)

Store CO₂ deep underground in geological formations for permanent trapping.

Carbonated Building Materials

Lock CO₂ into concrete or bricks, storing it for decades or centuries.

Reforestation / Biochar

Invest in natural carbon sinks to offset re-emitted CO₂.

